

Carola Bonamico

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EDUCATION

Politecnico di Torino

MSc in Computer Engineering – Software

Turin, IT

Sep 2024 – Expected Oct 2026

- GPA: 27.8/30
- Thesis: “From Biosignals to Words: Exploiting Novel Deep Learning Architectures for Speech Understanding”
- Main courses: Data Science and Database Technology, Software Engineering I/II, Web Applications I, Large Language Models for Software Engineering, Human Computer Interaction, Computer Architectures, System and Device Programming

University of Pavia

BSc in Biomedical Engineering

Pavia, IT

Sep 2020 – Apr 2024

- Thesis: “Literature review on the impact of sleep quality on glycemic patterns in patients with type 1 diabetes”
- Main courses: Biomedical Data Processing, Models of Biological Systems

I.I.S. Cellini – Liceo Linguistico Alberti

Secondary School Diploma in Modern Languages

Valenza, IT

Sep 2015 – Jun 2020

- Languages: English, French, German

EXPERIENCE

Huawei Seeds for the Future Italy 2026

Program Participant

Beijing, Shanghai, CN

Sep 2026 – Sep 2026

- Selected as one of 6 students out of over 100 applicants for Huawei’s fully-funded flagship talent program in Beijing and Shanghai.
- Engaged in a full-immersion training on technology, innovation, and sustainability, designed to combine academic merit and entrepreneurial spirit within the advanced Chinese ecosystem.

ETH Zurich

Master’s Thesis Student

Zurich, CH

Mar 2026 – Aug 2026

- Grade: 5.5/6.0
- Conducted Master’s thesis project at the Integrated Systems Laboratory (IIS) under supervision of Prof. Dr. Luca Benini, with focus on developing edge-constrained Deep Learning models for silent speech recognition from electromyographic signals.
- Reached 85.8% and 82.1% accuracy (vocalised, silent) in classification and 16.3% and 22.4% Word Error Rate (WER) (vocalised, silent) in continuous recognition.
- Designed and collected a multimodal, multi-session dataset from 7 subjects for 40 hours of synchronised EMG, EEG and audio.

Squadra Corse PoliTo

Software Developer

Turin, IT

Aug 2025 – Nov 2025

- Built a TUI in Go to send, receive and inspect CAN messages over the SocketCAN interface.

TECHNICAL SKILLS

Programming languages: C, Rust, Go, JavaScript, TypeScript, Python, ARM Assembly, MATLAB

Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn

Libraries and frameworks: Node.js, React.js, React Native, TypeORM, Diesel

Databases: Oracle SQL, MongoDB, MySQL, SQLite, PostgreSQL

Tools and software: Git, GitHub, GitLab, Docker, Jupyter Notebook, LabVIEW

Other: REST APIs, Agile, UML, HTML, CSS

PROJECTS

Autostar Motorsport Web App

Responsive web application for a motorsport company, built with Node.js and React.js. Implements a client-side architecture in JavaScript, GDPR compliance through Iubenda and modular React components styled with CSS and AOS.

SilentWear

Fork of SilentWear, a wearable system based on a textile neckband for silent speech recognition from EMG. Extended the existing CNN with a BiLSTM and a Transformer, added CTC decoding for direct EMG-to-text recognition alongside closed-set classification, and implemented an EMG-based automatic speech onset detection algorithm.

Participium

Full-stack platform for civic issue reporting in TypeScript/JavaScript (Node.js and web frontend), with role-based flows, geolocation and photo upload. Includes Telegram notifications and SonarQube for static analysis and continuous code quality monitoring.

Rust Chat

Real-time group messaging application built with Rust and React. Supports private chats, groups, JWT authentication, WebSocket communication and Docker deployment.

Geocontrol

Dockerised software monitoring system for managing geological, meteorological and environmental sensors. Built in TypeScript using the repository pattern, TypeORM and Node.js.

CERTIFICATIONS

Goethe-Zertifikat B1 (German) – Goethe-Institut Italia

Issued: Jul 2019

B1 Preliminary (English) – Cambridge English

Issued: May 2018

Delf A2 (French) – Ministère de l’Éducation nationale

Issued: Sep 2015